Introduction To Flight Mechanics & Control

Tutorial 6



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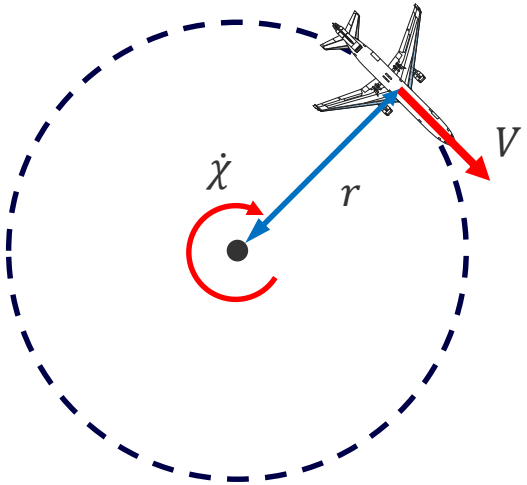
Key Facts

A turbojet aircraft shows following characteristics:

Aircraft mass	m	2000 kg
Drag factor	k	0,1
Reference wing area	S_{ref}	20 m^2
Drag coefficient	C_{D0}	0.023
Maximum lift coefficient	$C_{L,max}$	1
Maximum load factor	$n_{z,max}$	4
Thrust density dependency exponent	n_ρ	0.75
Maximum thrust at MSL	$T_{max,S}$	$0.55 \cdot m \cdot g$

Exercise 6.1

How high is the **minimum turn radius** (the turn radius where $C_{L,max}$ and $n_{z,max}$ are fulfilled) and the **corresponding aircraft speed at the height h = 0 m**? (the aircraft is supposed to fly horizontally)

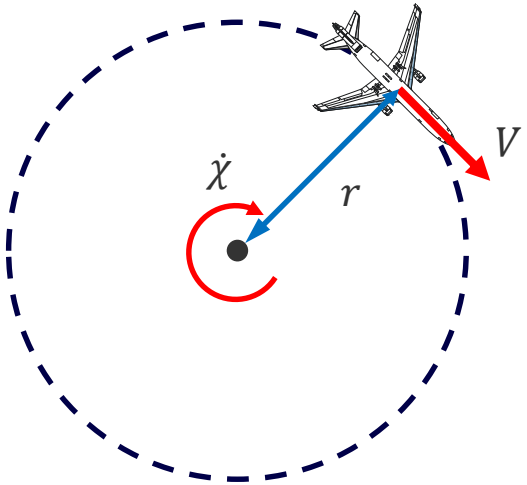


How high is the **minimum turn radius** (the turn radius where $C_{L,max}$ and $n_{z,max}$ are fulfilled) and the **corresponding aircraft speed at the height h = 0 m**? (the aircraft is supposed to fly horizontally)

General turn radius:

Kinematic relation

$$r_t = \frac{V}{\dot{\chi}}$$



Turn rate

$$\dot{\chi} = \frac{g}{V}\sqrt{n_z^2 - 1}$$

Result of PM-EoM consideration (more information on the following recap slides)

Using the definition of turn in the point mass equations of motion

$$\dot{V} = \frac{T - D}{m} - g\underbrace{\sin\gamma}_{=0} = 0 \qquad \dot{\chi} = \frac{L\sin\mu}{m\,V\underbrace{\cos\gamma}_{=1}} \qquad \dot{\gamma} = \frac{L\cos\mu}{m\,V} - \frac{g}{V}\underbrace{\cos\gamma}_{=1} = 0$$

$$\rightarrow T = D \quad \Leftarrow \text{Horizontal, Stationary}$$

$$\rightarrow L\sin\mu = mV\dot{\chi} \quad \Leftarrow \text{Horizontal, Stationary}$$

$$\rightarrow L\cos\mu = m\,g \quad \Leftarrow \text{Horizontal, Stationary}$$

$$\frac{L\sin\mu}{L\cos\mu} = \frac{m\,V\,\dot{\chi}}{m\,g} \rightarrow \tan\mu = \frac{V\,\dot{\chi}}{g}$$

$$L^2\left(\sin^2\mu + \cos^2\mu\right) = (m\,V\dot{\chi})^2 + (m\,g)^2 \rightarrow L = m\,g\sqrt{1 + \left(\frac{V\,\dot{\chi}}{g}\right)^2}$$

Question: Is there a parameter defining the turn performance of an aircraft?

Introduction of the loadfactor n_z

For defining turn performance, we introduce the load factor n_z:

$$(n_z)_K = \frac{L}{m\,g}$$

Acceleration due to lift

Normalization with respect to weight

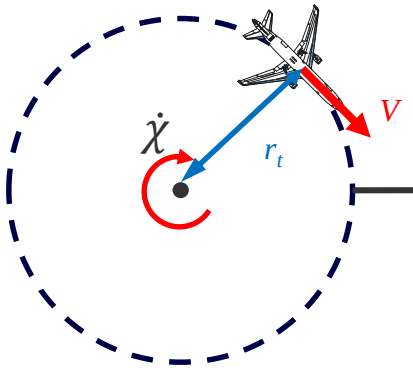
The load factor is the lift force normalized by the gravitational force. This normalization leads to $n_z = 1$ in straight horizontal flight.

$$\left(n_{z,Horizontal}\right)_K = 1$$

With this definition of the load factor, the available turn performance can be described, as the turn performance results from an excess of lift over weight.

Annotation: This interpretation of load factor is fit to the following turn performance. However, the benefit and usage of the load factor can be applied to more problems as well.

How can the turn radius be obtained?



The turn radius depends on aircraft velocity and turn rate

$$r_t = \frac{V}{\dot{\chi}}$$

Goal: dependency between turn rate, turn radius and n_z, in order to answer the question of minimum possible turn rate

$$n_z = \frac{L}{m\,g} \xrightarrow{L = m\,g\sqrt{1+\left(\frac{V\,\dot{\chi}}{g}\right)^2}} n_z = \sqrt{1+\left(\frac{V\,\dot{\chi}}{g}\right)^2} \xrightarrow{\dot{\chi} = \frac{V}{r_t}} n_z = \sqrt{1+\left(\frac{V^2}{g\,r_t}\right)^2}$$

$$n_z = \frac{L}{m\,g} \rightarrow n_z = \frac{1}{\cos\mu}$$

$$n_z = \sqrt{1+\left(\frac{V\,\dot{\chi}}{g}\right)^2} \xrightarrow{\tan\mu = \frac{V\,\dot{\chi}}{g}} \tan\mu = \frac{V\,\dot{\chi}}{g} \xrightarrow{\text{Solving for } \dot{\chi}} \dot{\chi} = \frac{g}{V}\tan\mu \xrightarrow{n_z = \sqrt{1+\tan^2\mu} \rightarrow \tan\mu = \sqrt{n_z^2-1}} \dot{\chi} = \frac{g}{V}\sqrt{n_z^2-1}$$

Minimum Turn Rate

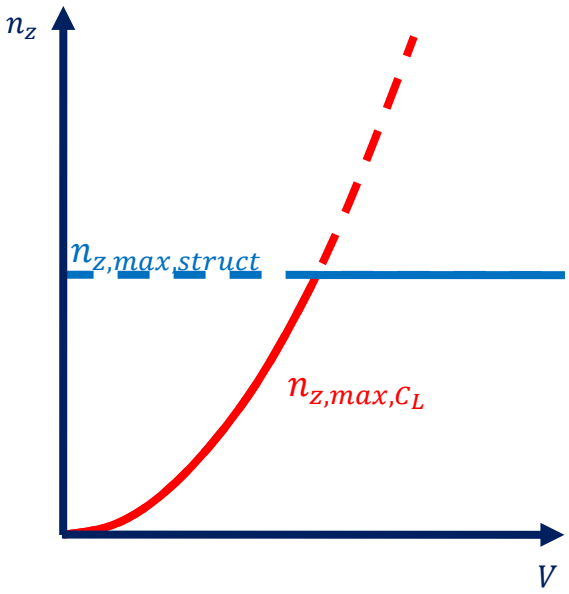
$$\dot{\chi} = \frac{g}{V}\sqrt{n_z^2 - 1} \qquad r_t = \frac{V}{\dot{\chi}} \qquad r_t = \frac{V^2}{g\sqrt{n_z^2 - 1}}$$

We quickly see that the minimal turn radius is achieved if the (squared) load factor n_z^2 is maximal!

The load factor (and thus the minimum turn radius) is physically limited by:

n_{z,max,C_L} → Aerodynamic limit

$n_{z,max,struct}$ → Structural limit



Minimum turn radius limitations in more detail

1. Due to structural load:

$$r_{t,min,Struct} = \frac{V^2}{g\sqrt{\boldsymbol{n_{z,maxStruct}^2} - 1}}$$

already given

2. Due to aeordynamics:

$$r_{t,min,CL} = \frac{V^2}{g\sqrt{\boldsymbol{n_{z,maxCL}^2} - 1}} = \frac{V^2}{g\sqrt{\left(\frac{\frac{1}{2}\cdot\rho\cdot V^2\cdot S_{ref}\cdot C_{Lmax}}{m\cdot g}\right)^2 - 1}}$$

With the load factor at maximum lift $n_{z,max,CL}$:

$$n_{z,max,CL} = \frac{\frac{1}{2}\cdot\rho\cdot V^2\cdot S_{ref}\cdot C_{Lmax}}{m\cdot g}$$ = max lift/mg

Minimum turn radius limitations in more detail (cont'd)

1. Due to structural load:

$$r_{t,min,Struct} = \frac{V^2}{g\sqrt{\boldsymbol{n_{z,maxStruct}^2} - 1}}$$

Turning is "harder" with increasing velocity…

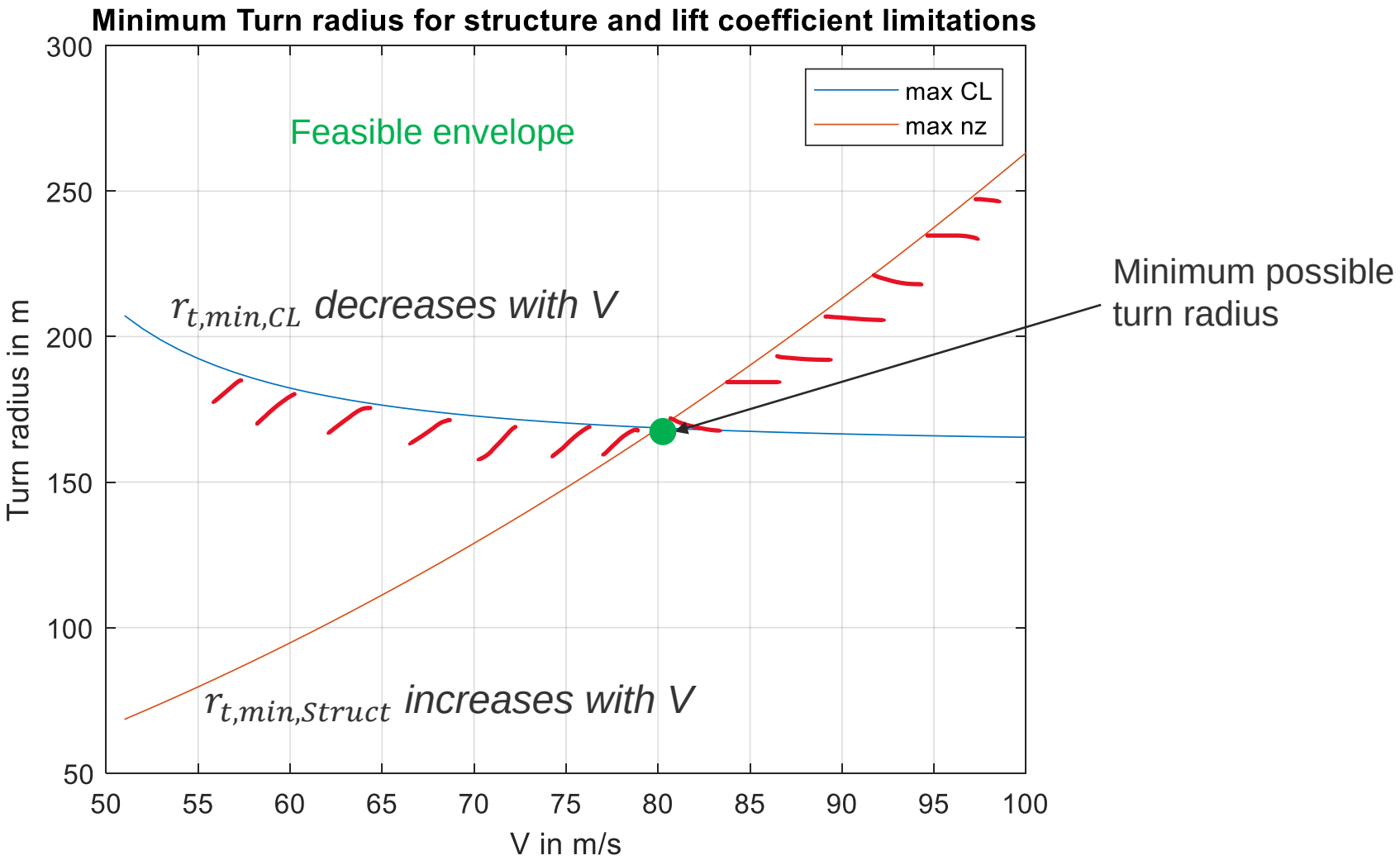
2. Due to aeordynamics:

$$r_{t,min,CL} = \frac{V^2}{g\sqrt{\boldsymbol{n_{z,maxCL}^2} - 1}} = \frac{V^2}{g\sqrt{\left(\frac{\frac{1}{2}\cdot\rho\cdot V^2\cdot S_{ref}\cdot C_{Lmax}}{m\cdot g}\right)^2 - 1}}$$

With more lift we can turn „better"…

→ How does the minimum turn radius change with velocity?

We observe two opposing effects!



Minimum turn radius at the intersection of both constraints

$$r_{t,min} = \frac{V^2}{g\sqrt{n_{z,maxStruct}^2 - 1}} = \frac{V^2}{g\sqrt{n_{z,max,CL}^2 - 1}} = \frac{V^2}{g\sqrt{\left(\frac{\frac{1}{2}\cdot\rho\cdot V^2\cdot S_{ref}\cdot C_{Lmax}}{m\cdot g}\right)^2 - 1}}$$

Remark, both equations only hold for loadfactor bigger than 1!!

$$n_z = \frac{L}{m\cdot g}$$

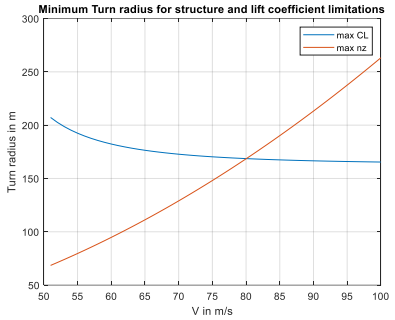
→ Sustain the turn, i.e. horizontal flight!

Therefore, lift must be higher than weight → only holds for velocities higher than the **minimum velocity for horizontal flight**!

$$V_{min} = \sqrt{\left(\frac{2\cdot m\cdot g}{\rho\cdot C_{Lmax}\cdot S_{ref}}\right)}$$

Minimum turn radius at the intersection of both constraints

$$r_{t,min} = \frac{V^2}{g\sqrt{n_{z,maxStruct}^2 - 1}} = \frac{V^2}{g\sqrt{n_{z,max,CL}^2 - 1}} = \frac{V^2}{g\sqrt{\left(\frac{\frac{1}{2}\cdot\rho\cdot V^2\cdot S_{ref}\cdot C_{Lmax}}{m\cdot g}\right)^2 - 1}}$$



$$\frac{\rho\cdot V^2\cdot S_{ref}\cdot C_{Lmax}}{2\cdot m\cdot g} = n_{zmax,struct}$$

$$V^2 = \frac{2\cdot n_{zmax,struct}\cdot m\cdot g}{\rho\cdot C_{Lmax}\cdot S_{ref}} \rightarrow V_{r_{min}} = \sqrt{\frac{2\cdot 4\cdot 2000\cdot 9{,}81}{1{,}225\cdot 1\cdot 20}}\frac{m}{s} = 80{,}04\frac{m}{s}$$

$$r_t = \frac{V^2}{g\sqrt{n_{z,max,struct}^2 - 1}} = \frac{80{,}04^2}{9{,}81\cdot\sqrt{4^2-1}}m = 168{,}62m$$

Can this turn radius be achieved from a propulsion point of view?
Hint: How high is the required thrust?

Horizontal flight: $\boxed{T = D}$ ← no flight path angle

where,

$$D = \frac{1}{2} \cdot \rho \cdot V^2 \cdot C_D \cdot S_{ref} = \frac{1}{2} \cdot \rho \cdot V^2 \cdot \left(C_{D0} + k \cdot C_L^2\right) \cdot S_{ref}$$

$$\Rightarrow \quad T_{req} = \frac{1}{2} \cdot \rho \cdot V^2 \cdot S_{ref} \cdot \left(C_{D0} + k \cdot C_L^2\right)$$

We know that we must fly with C_{Lmax} in order to achieve the minimum turn radius:

$$\boxed{T_{req} = \frac{1}{2} \cdot \rho \cdot {V_{r_{min}}}^2 \cdot S_{ref} \cdot \left(C_{D0} + k \cdot C_{L,max}^2\right)}$$

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$$T_{req} = \frac{1}{2} \cdot 1{,}225 \cdot 80{,}04^2 \cdot 20 \cdot \left(0{,}023 + 0{,}1 * 1^2\right) N = 9652{,}85\ N$$

Let's compare this to the available thrust:

$$T_{max} = 0{,}55 \cdot m \cdot g = 10791{,}00\ N$$

Exercise 6.3

From what **height** on does the **available thrust determine the possible minimum turn radius**? How high is the **turn radius at this altitude**? Use the standard atmosphere.

From what **height** on does the **available thrust determine the possible minimum turn radius**? How high is the **turn radius at this altitude**? Use the standard atmosphere.

What is the influence of the altitude?

$C_{L,max}$ and $n_{z,max,Struct}$ do not depend on the altitude

Does the required thrust change?

Required thrust: $$T_{req} = \frac{1}{2} \cdot \rho \cdot V^2 \cdot S_{ref} \cdot \left(C_{D0} + k \cdot C_{L,max}^2\right)$$

As there is the density in the equation, we could assume that the required thrust depends on the height?

But alternative view shows:

$$T = D = \frac{D}{L} \cdot L = \frac{C_D}{C_L} \cdot n_z \cdot m \cdot g$$

For this exercise:

$$T_{req} = D = \frac{D}{L_{max}} \cdot L_{max} = \frac{C_{D,max}}{C_{L,max}} \cdot n_{z,max} \cdot m \cdot g$$

None of the variables do depend on the density.
→ Required thrust is not depending on altitude!

Question: Why is the density in the slide before?
→ it cancels out with V^2!

For minimum radius: $V^2 \sim \frac{1}{\rho}$ $V^2 = \frac{2 \cdot n_{zmax,Struct} \cdot m \cdot g}{\rho \cdot C_{Lmax} \cdot S_{ref}}$

However, recall that the available thrust changes with the altitude (density)!

$$\frac{T}{T_i} = \left(\frac{V}{V_i}\right)^{n_V} \cdot \left(\frac{\rho}{\rho_i}\right)^{n_\rho}$$

Turbojet aircraft: $n_v = 0$ → no change in thrust with velocity

There must be an altitude from which on the producible thrust is below the (constant w.r.t. ρ) required thrust.

$$\Rightarrow \quad \rho_{crit} = \left[\frac{T_{req}}{T_{max,S}}\right]^{\frac{1}{n_\rho}} \cdot \rho_{S,msl} = 1.0559 \frac{kg}{m^3}$$

At this point we can compute the density where our available thrust equals the required thrust:

$$\rho_{crit} = \left[\frac{T_{req}}{T_{max,S}}\right]^{\frac{1}{n_\rho}} \cdot \rho_{S,msl} = 1.0559\,\frac{kg}{m^3}$$

Using the standard atmosphere, we can then easily compute the altitude:

$$\Rightarrow \quad \boxed{h_{geod} = \frac{r_{earth} \cdot H_G}{r_{earth} - H_G} = 1521.01\,m}$$

Where:

$$H_G = \frac{\left(\left(\frac{\rho_{crit}}{\rho_{S,MSL}}\right)^{n-1} - 1\right)}{\gamma_{TR}} \cdot T_{S,MSL} = 1520.65\,m$$

From what **height** on does the **available thrust determine the possible minimum turn radius**? How high is the **turn radius at this altitude**? Use the standard atmosphere.

Finally, we want to compute the **turn radius at this "maximum" altitude**:

$$r_t = \frac{V^2}{g\sqrt{n_{z,max,Struct}^2 - 1}}$$

To this end we must first compute the corresponding velocity. We know that the density is inversely proportional to the velocity, with which we fly the turn. (slide 15)

For minimum radius: $V^2 \sim \frac{1}{\rho}$ $V^2 = \frac{2 \cdot n_{zmax,Struct} \cdot m \cdot g}{\rho \cdot C_{Lmax} \cdot S_{ref}}$

$$\frac{V_S^2}{V^2} = \frac{\rho}{\rho_S} \rightarrow V = \sqrt{\frac{\rho_S}{\rho_{crit}}} \cdot V_S = \sqrt{\frac{1{,}225}{1.0559}} \cdot 80{,}04 \frac{m}{s} = 86{,}21 \frac{m}{s}$$

From what **height** on does the **available thrust determine the possible minimum turn radius**? How high is the **turn radius at this altitude**? Use the standard atmosphere.

Finally, we want to compute the turn radius at this “maximum” altitude:

$$r_t = \frac{V^2}{g\sqrt{n_{z,max,Struct}^2 - 1}}$$

$$\Rightarrow \quad \boxed{r_t = \frac{86{,}21^2}{9{,}81\sqrt{4^2 - 1}}\, m = 195{,}63\, m}$$